Validation report

Pewman Innovation

15.03.2024

Validation ID: RM0029

Impact Forecast

Impact Insights & additionality

Contents

Details of the validation process

Timestamps and results:
The validation documented in this report was delivered with the following time stamps and results:

Pewman Innovation	Validation request	First review	Feedback call	Hand-in revisions	Final review	Wrap-up call
Date	31/01/25 15h13	03/02/25 21h06	21/02/25 14h00	10/03/25 16h36	12/03/25 15h46	14/03/25 15h30
Result	Plausible, positive and significant			Valid, positive and significant		

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Author	Rafael Bartolomeu Martins
Company name	Pewman Innovation
Project CIF lead	Enzo Galliani, Operations director/Cofounder
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Introduction to CIF Validation

To determine the validity of self-assessed climate impact forecasts we provide CIF Validation, which is a third party verification of the calculation of the climate and environmental impact of an innovation, in order to conclude if the Climate Impact Forecast is valid, positive and significant.

Problem solved

There are areas of LCA expertise that can not be covered in the Climate Impact Forecast workshops or CIF Training, for example where domain knowledge and experience are required. With self-assessments there is also a risk of optimism bias. Validation assures that forecasts do not contain gaps, scoping errors, unsupported assumptions or inappropriate data sources. CIF Validations are made on the request of the project team, and possibly commissioned by an impact organisation. The results are used by teams and organisations to compare and communicate the climate impact of projects.

A validation process performed by an impartial impact expert, who has read about the innovation, seen the forecast and used a checklist to assess its validity. The validator provides detailed written feedback and offers the opportunity for a revision. The goal of this process is twofold: increase the quality of a forecast and to conclude if the forecast is suitable to draw conclusions about the positive climate impact of the innovation. This Validation report documents the results of that process.

Definitions of key terminology

Climate Impact Forecast (CIF)	A Climate Impact Forecast or CIF is an LCA based calculation of the GHG reduction or climate adaptation potential of a project. Using our CIF tool, the project team found the net climate impact of the key differences between business as usual and their innovative solution.
CIF Validation process	A review process delivered by a validator and guided by a structured check of the information entered into a CIF, a sensitivity analysis and the write-up of an Impact story. This process usually takes two weeks and includes a first review, a first feedback call between the team and validator, time for revisions if needed, a final review and a final results call.
Validator	Validations are delivered by Validators; CIF trainers with LCA expertise who are trained to perform this process in a uniform and objective way. Other than providing this service, Validators have no relationship with or obligations to the company or supporting organisation requesting the validation, assuring an impartial third party review.
Validation result	The CIF Validation result consists of three independent outcomes, which in the best case are valid, positive and significant. These qualifications and the alternative outcomes are explained on the next page.

The CIF Validation result consists of three independent outcomes

Validity of the forecast

A CIF is valid if it is representative of the project, using appropriate data and well-justified assumptions. Therefore, the CIF and its results are representative of the potential for the project to mitigate, enable or adapt to climate change.

Detailed requirements for validity are specified on www.impact-forecast.com/ CIF-validations. A CIF can be:

Valid	Plausible	Improbable	Invalid

Reduction potential

A CIF is positive when it shows that the project has a lower climate impact than business as usual, or improved climate resilience in the case of adaptation. A positive mitigation or enabler CIF file shows the avoided GHG emissions in $-tCO_2eq$.

This outcome depends on a sensitivity assessment. CIF results can be:

Positive	Positive within limits	Unclear	Sensitive	Negative

Impact threshold

A CIF is significant when the project has a climate impact (positive or negative) greater than 5 tonnes of CO_2eq per year. This is roughly the global average annual CO_2 emissions per person and the mass of a male African Elephant.

The threshold for significant impact can be set to a higher amount for a particular organisation or occasion. The result can be:

Significant	Marginal

Pewman Innovation CIF Validation

This validation consists of the following sections

Impact story	An impact story is a summary of how a project makes a positive climate impact. It is written by the validating impact expert and contains the key impact data from the Climate Impact Forecast.
Climate Impact Forecast and Validation result	The Climate Impact Forecast shows the scope and parameters of the impact calculation. This includes the resources used and saved by the innovation, their amount and climate impact, the climate impact per unit of user, and the total climate and environmental impact for all units or users in the timeframe. Validator feedback is included on strong and weak points of the forecast as a whole, as well as the conclusion from the sensitivity assessment and the approval status of individual parameters. The conclusion of the validation process is noted in the Validation result.
Sources and assumptions	The differences (resources used and reduced by the innovation, compared to the baseline solution) and quantities (of materials, energy etc.) in the forecast are based on sources and assumptions specified in this section.

Impact story

Rethinking Frost Protection: Biofortifier vs. Anti-Frost Towers

Pewman Innovation introduces Crioprotect, an anti-frost biofortifier designed to safeguard crops sustainably. Utilising a triple-action mechanism that combines bionanopolymers, Antarctic bacteria, and organic compounds, Crioprotect offers an alternative to anti-frost towers, which rely heavily on fossil fuels. Crioprotect provides a low-carbon solution for

The climate impact assessment compares Crioprotect against the baseline use of anti-frost towers, evaluating the annual difference in emissions. The total impact is measured based on the protection of 1,000 hectares per year, providing a comprehensive view of its environmental benefits, leading to a reduction in the order of 1500 tonnes of CO_2eq.

How does this make a positive climate impact? Compared to which baseline?

Crioprotect offers a positive climate impact by replacing traditional anti-frost towers, which are energy-intensive and rely on fossil fuels for operation, with biofortified. Unlike anti-frost towers, which generate significant CO_2 emissions during both production and operation, Crioprotect uses biological ingredients like bionanopolymers, Antarctic bacteria, and organic compounds, all of which have a much lower environmental footprint.

The baseline for comparison is the anti-frost towers, which have been widely used in agriculture for over 100 years, especially in crops like grapes, citrus, and cherries. While effective at preventing frost damage, these towers consume large amounts of diesel fuel and have high maintenance costs, contributing to a higher carbon footprint.

How much of a climate impact, and what does the impact depend on?

The climate impact of Crioprotect is measured by the reduction in CO_2 emissions compared to the baseline of anti-frost towers. The main impact driver is the potential that it has in reducing diesel generation required by the anti-frost towers.

- Product Usage: The amount of Crioprotect used per hectare, which is around 30 liters per hectare annually.
- Energy Use: The electricity required to produce the product, which is much lower than the energy consumed by operating anti-frost towers.
- Transport: The distance and type of transport needed for Crioprotect versus anti-frost towers, including water for dilution and the tractor energy for applying the product.
- Materials: The raw materials used to create Crioprotect and the environmental impact of producing and shipping the product.

When compared to anti-frost towers, the primary reduction in impact arises from the elimination of diesel fuel usage and a sustainable product lifecycle. Given that the baseline uses 402 kg of diesel annually, Crioprotect has the potential to reduce CO_2 equivalent emissions by 1,456 kg through its avoidance.

Validity

The forecast is valid, positive and significant.

It is valid as all checks have been approved, with no questions or concerns remaining. Any external information has been verified, and the impact is robust. A more detailed Life Cycle Assessment (LCA) is expected to yield results within the same range.

It is considered positive considering that the forecast shows that the innovation robustly reduces CO_2 impact. When the primary impact driver - diesel reduction - is reduced to zero quantities, the innovation consistently leads to a significant reduction in CO_2 impact, approximately -18 tonnes per protected hectare.

Finally, it is considered Significant because it has the potential impact greater than 5 tons per year, which compensates for more than one average person. The total positive climate impact expressed in an amount that is meaningful in this context.

Climate Impact Forecast and Validation result

Pewman Innovation provides Crioprotect, an anti-frost biofortifier with a triple-action mechanism using bionanopolymers, Antarctic bacteria, and organic compounds, capable of protecting crops on any scale instead of the baseline solution for frost protection that typically involves anti-frost towers, which, while effective, consume large amounts of fossil fuels and come with high maintenance costs. The difference in impact is calculated per year and the total impact of Pewman Innovation per year is calculated for 1000 times 1 protected hectare.

Validation	By: Rafael Martins, Started: Wed Mar 12 2025 15:32:08 GMT+0100 (Central European Standard Time), Completed: Wed Mar 12 2025 15:46:01 GMT+0100 (Central European Standard Time)
Strong points	(+) Clear scope statement, making it clear how the innovation replaces the baselinte (+) Proper and descriptive differences and assumptions, supporting them with resources where need (+) Clear calculation
Sensitivity	When reducing the main impact driver to null quantities, the innovation robustly reduces CO2 impact, in the order of -18 tonnes per protected hectare.

Production

+	Electricity Chile	0.1123 per MJ	0,144	kWh	0.05821
+	drinking water europe	0.00063 per kg	30	kg	0.01887
+	Amino acids, peptides, and analogues (n=1	20.23 per kg	0,0003	kg	0.00607
+	Sodium Chloride, NaCl (plasticseurope)	0.06 per kg	0,0003	kg	0.00002
+	Yeast extract	3,204 per kg	0,00015	kg	0.00048
+	PET (Polyethylene terephthalate) chemical	1.263 per kg	1,357	kg	1.714
+	Glycerol	2 per kg	0,000075	kg	0.00015
−	Stainless Steel (secondary), average	1.935 per kg	10,5975	kg	-20.51
−	Hot-dip Zinc plating, pieces, outside use	6.953 per m2	0,1712	m2	-1.19

Transport

+	Truck+trailer 24 tons net (min weight/volur	0.08871 per tkm	3	tkm	0.2661
−	Container ship (min weight/volume ratio 0,	0.00478 per tkm	160,65	tkm	-0.7672
−	Truck+trailer 24 tons net (min weight/volur	0.08871 per tkm	1	tkm	-0.08871

Use

+	drinking water europe	0.00063 per kg	2970	kg	1.869
+	Tractor (240 pk)	0.2209 per tkm	3	tkm	0.6628
−	Diesel low-sulphur including combustion C	3.622 per kg	402	kg	-1456

CIF model continues on next page

Pewman Innovation's total impact per year

eco-costs of human health euro	-8126 + ?
eco-costs of eco-toxicity euro	-20233 + ?
eco-costs of resource depletion	-173904 + ?
eco-costs of carbon footprint	-196047 + ?

	Carbon footprint CO_2eq.
Impact per 1 protected hectare	-1474 kg
Impact of 1000 times 1 protected hectare	-1.5Kt

Validation ID:
RM0029
Date:
13-03-2025

Pewman Innovation
Mitigation project

Impact reduction potential	-1.5 KtCO_2eq./year

Validity of the forecast	Valid
Reduction potential	Positive
Impact threshold	Significant

Sources and assumptions

The differences and quantities in the forecast are based on the following sources and assumptions:

Production

The average product usage depends on the crop we aim to protect. For instance, vegetables require around 4 liters per hectare, while larger trees need approximately 40 liters per hectare. For this analysis, we chose cherry crops because they are our primary clients, concentrating 87% of the total volume of the product we sold in 2024. This percentage represents approximately 30 tons of product, which are used on roughly 1000 hectares. In our calculations, we compared CRIOPROTECT with the most commonly used baseline in these plantations in Chile: anti-frost towers. This solution has been used for more than 100 years now, mainly in grapes, citrus and deciduous trees, like cherries (https://www.fao.org/4/y7223e/y7223e0d.htm). On average, our cherry crop clients use 30 liters of product per hectare per year. In contrast, anti-frost towers cover 4 hectares each, equating to 0,25 towers per hectare per year.

The calculations are presented as follow:

0. Item -> Briefly explain how the item is used -> Justify or explain the reasoning behind the value -> The specific quantity or value of the item required annually to protect one hectare.

1. Electricity Chile -> For the production of the product, we cultivate 4 liters of concentrated bacteria in a bioreactor for 24 hours. The bioreactor model used is the Minifors 2 (Infors AG). According to the manual, under the Connection Requirements section, the maximum power consumption is listed as 800 W -> First, we calculated the energy required to produce 4 liters of concentrate by multiplying the maximum power consumption by the hours of use. This value was then divided by 4 and subsequently by 1000 (since our product concentration is 0,1%) to determine the energy needed for just 1 liter of the final product. Finally, the resulting number was multiplied by 30 to calculate the energy used to protect 1 hectare -> 0,144 kWh

2. Drinking water -> For producing 1 liter of product we use a liter of water. -> 30 kg

3. Peptides -> For preparing the medium to grow the bacteria, we use tryptone. We assumed it to be a generic peptide since it is digested casein. -> For 1 liter of medium, we use 0,01 kg of tryptone, which means that for 1 liter of the final product, we use 0,00001 kilograms of tryptone. To protect 1 hectare, we multiply this value by 30 -> 0,0003 kg

4. Sodium Chloride -> For preparing the medium to grow the bacteria, we use NaCl. -> For 1 liter of medium, we use 0,01 kg, which means that for 1 liter of the final product, we use 0,00001 kg of NaC l. To protect 1 hectare, we multiply this value by 30 -> 0,0003 kg

5. Yeast extract -> For preparing the medium to grow the bacteria, we use yeast extract. Due to the lack of specific data, we assumed that the yeast extract has the same environmental impact in its life cycle assessment (LCA) as regular yeast.(https://www.aflevadura.com/sostenibilidad/huella-de-carbono) -> For 1 liter of medium, we use 0,005 kg, which means that for 1 liter of the final product, we use 0,000005 kg of yeast extract. To protect 1 hectare, we multiply this value by 30 -> 0,00015 kg

6. PET -> We use PET for our packaging, specifically for the containers. For 30 liters of product, one 20-liter container and two 5-liter containers can be used.-> We weighed the containers to calculate the total PET used to protect 1 hectare. -> 1,357 kg

7. Glycerol -> The impact of glycerol was not found in the database, so we referenced it (https://apps.carboncloud.com/climatehub/product-reports/id/16243501676). Glycerol is used to cultivate the bacteria. -> For 1 liter of bacteria, we use 0,0025 kg of glycerol, which means that for 1 liter of the final product, we use 0,0000025 kg. To protect 1 hectare, we multiply this value by 30 -> 0,000075 kg. To calculate the impact of the baseline, anti-frost towers, we considered a service life of 25 years (https://ag-group.es/torres-antiheladas/). All calculations were based on the annual usage of the tower. Additionally, we know that one tower protects approximately 4 hectares, so the calculations were adjusted to account for 0,25 towers being used to protect one hectare.

For determining the kg CO_2 eq generated during the production of anti-frost towers, we assumed that nearly all emissions come from the primary material, galvanized steel. This material consists of steel and a zinc coating. We referenced the dimensions of a three-blade anti-frost tower from AG Group (https://ag-group.es/torres-antiheladas/) and used a steel density of 7.850 kg/m³.(https://inoxidablesvictoria.com/blog/cual-es-la-densidad-del-acero-inoxidable/). However, we were unable to calculate the energy used during the fabrication process. Furthermore, components such as nails, bolts, and the materials used for the blades, among other elements, were not included in the calculations.

8. Stainless Steel -> This is the main material of the tower. We used the dimensions provided by the manufacturer: a tube 10,7 meters tall, with a thickness of 7,9 millimeters and a diameter of 0,51 meters. From these dimensions, we calculated the surface area and volume of the tube's sheet. The unfolded sheet would measure 10,7 meters in height, 1,6 meters in length, and 0,0079 meters in thickness. -> The calculated volume was 0,135 m³, and multiplying this by the density of steel gave us the total amount of steel used in one tower. One-quarter of a tower corresponds to 264,93 kg of steel. Dividing this by the service life of the tower, we obtain -> 10,5975 kg

9. Hot-dip Zinc Plating -> This coating is used to protect the towers from environmental factors and extend their service life. -> The calculated area for the sheet tube is 17,12 m². One-quarter of this surface area is used per hectare, and dividing it by the service life in years gives us -> 0,1712 m²

Transport

For the transport we consider an average client that is located 100 km from the production center.

1. Truck (Crioprotect) -> to protect one hectare per year we need to transport 30 kg (0,03 ton) of product though 100 km. -> 3 tkm

2. Container ship -> The towers are not produced in Chile, so they must be shipped by sea. In this case, we considered that the tower comes from New Zealand (https://www.frostboss.com/es-cl/about/who-we-are), and the distance a ship must travel to deliver the

tower is 16065 km (https://www.fluentcargo.com/routes/new-zealand/chile). Each tower weighs nearly one ton. -> We calculated the tkm, divided this number by 4, and then by the lifespan -> 160,65 tkm

3. Truck (anti-frost towers) -> To transport the tower to the final client, nearly a ton needs to be transported over a distance of 100 km. This number is then divided by 4 and further divided by the lifespan of the tower. -> 1tkm

Use

For the usage of the product, we calculated the water required to dilute it and included the tractor application in the calculations.

1. Drinking water -> For this crop, a product concentration of 1% is recommended, which means diluting 1 liter of product in 99 liters of water. For one year, nearly 3 tons of water are needed -> 2,970 tons

2. Tractor -> On average, the recommended application schedule per year includes 3 applications, each using 10 liters of product diluted in 990 liters of water. Additionally, we assumed a 100 x 100 m hectare with 20 rows of trees (a typical 5 x 2 plantation framework for cherry trees). This setup results in the tractor traveling 2 km per application, starting with a tank weighing 1 ton and ending at zero tons, repeated three times. -> The sum of the areas under the curve for the 3 applications is -> 3 tkm For the operation of the tower, we considered the amount of diesel used.

3. Diesel -> Diesel is required for the towers to operate. On average, a tower (with the smallest motor CAT 3.4 Tier 3) consumes 18.93 liters of diesel per hour, according to the manufacturer (https://ag-group.es/torres-antiheladas/). The manufacturer also estimates that the towers are used for approximately 100 hours per year. -> To calculate the fuel used to protect one hectare, we first multiplied the average liters per hour by the hours of use, then divided the result by 4, and finally multiplied it by the density of diesel (850 g/L). -> 383 kg/year.

More information

For more information about this validation, and Climate Impact Forecast Validation in general, reach out to Impact Forecast.

Impact Forecast B.V.
The Netherlands
info@impact-forecast.com